DECIPHERING DIPLOPIA

DR RAMYA R NADIG

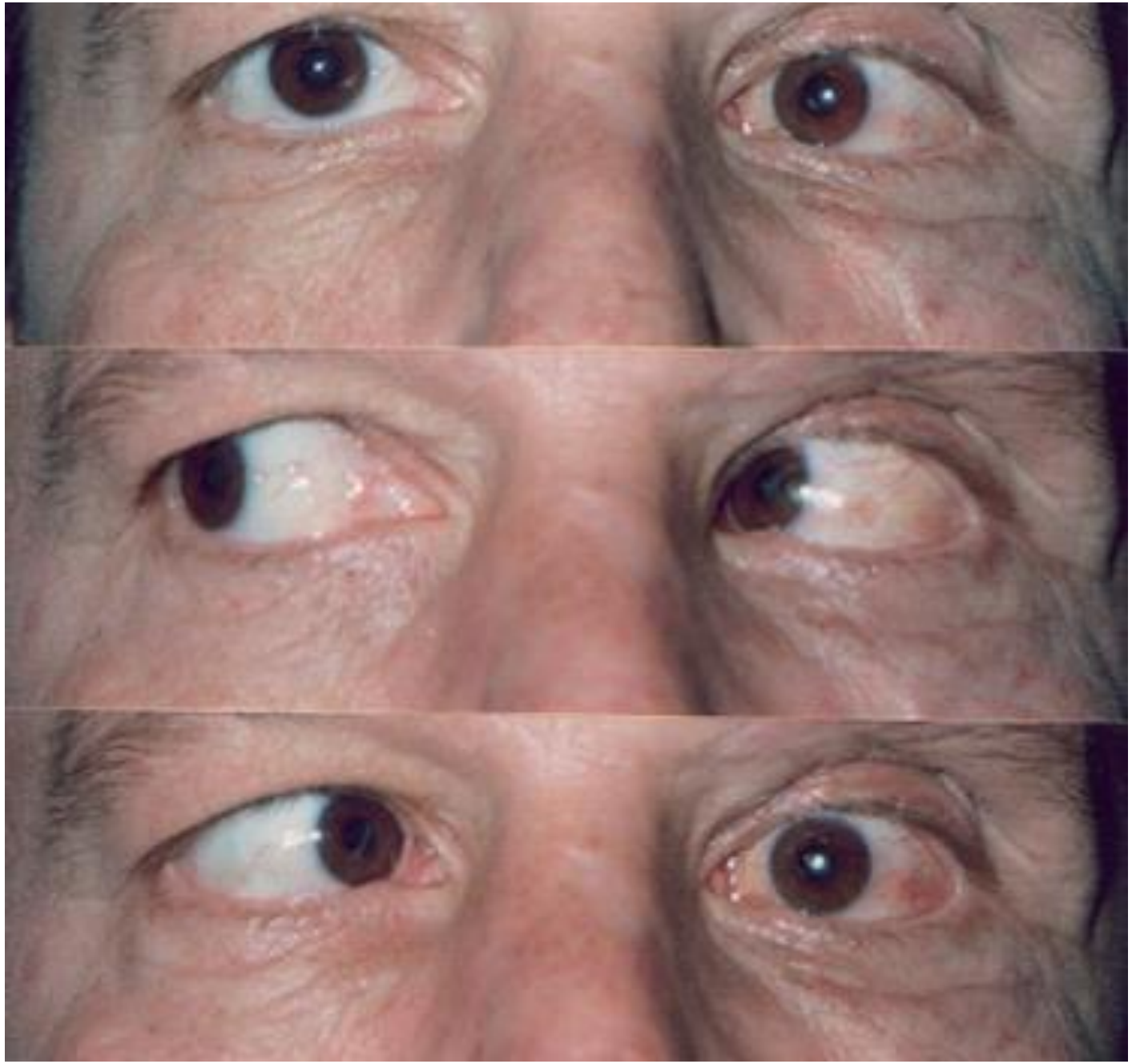
VIJAY BALAJI K

DOCTOR, I SEE DOUBLE !!



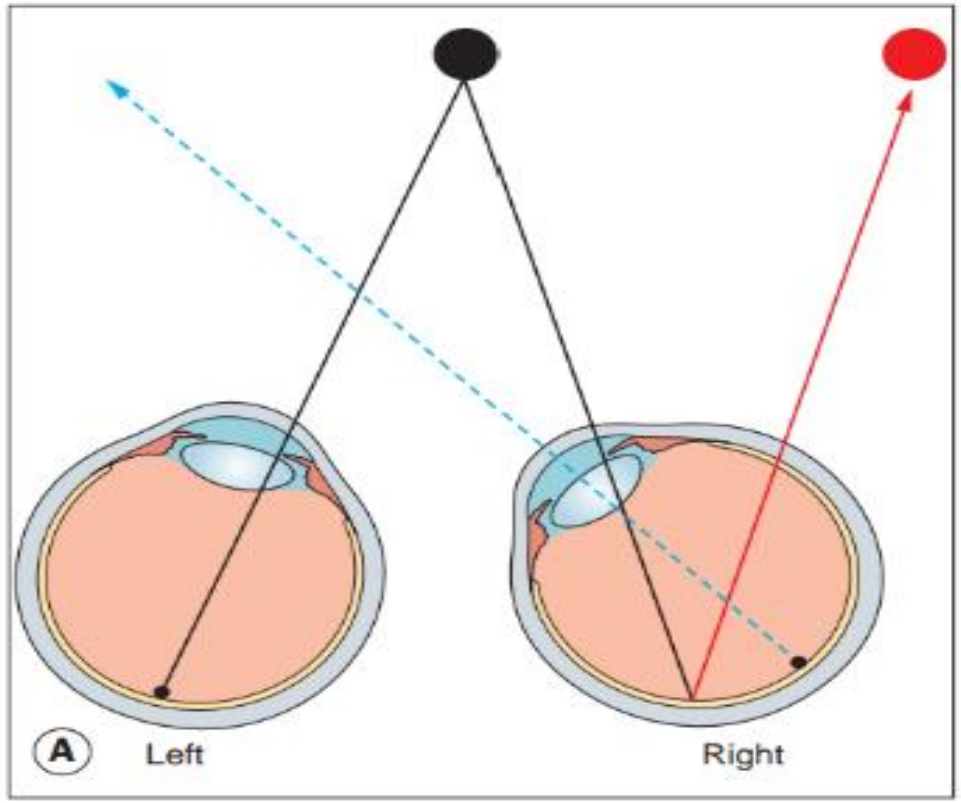
CASE 1:

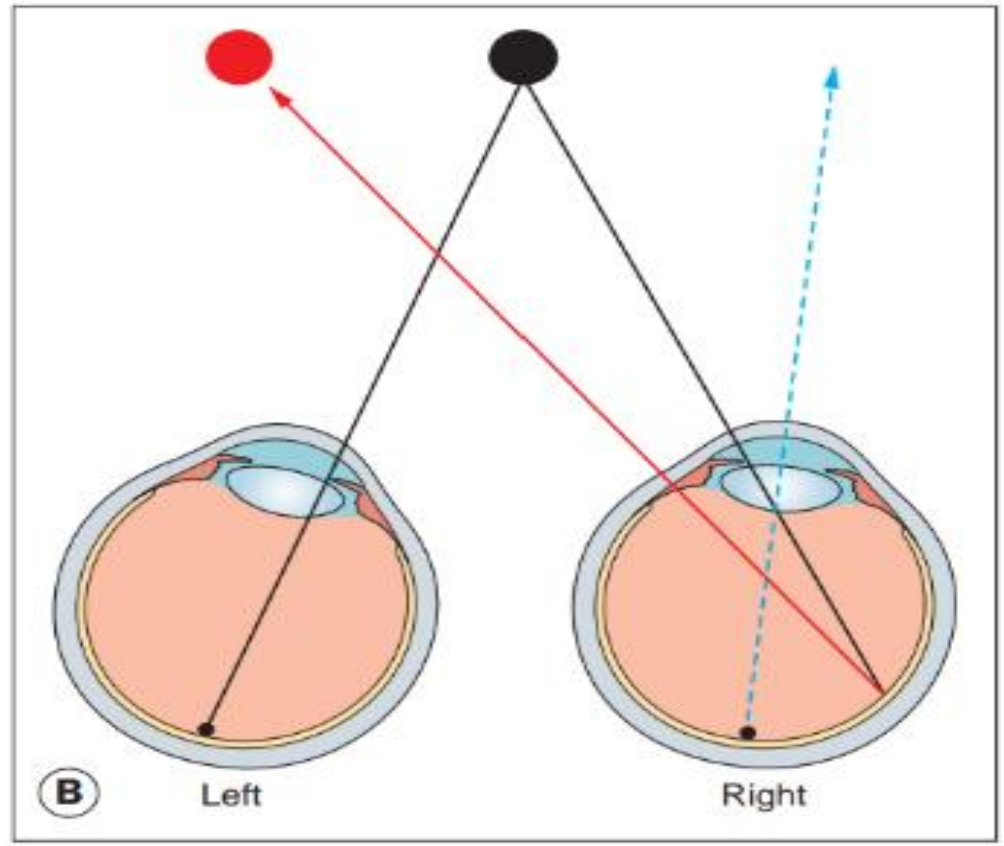
- A 64-year-old man presented for evaluation of binocular horizontal diplopia that began 2 weeks earlier and was most bothersome when viewing at a distance and when driving. He noted that the diplopia was more prominent when he directed his gaze to the left but otherwise it was present“all the time.” He had a history of hypertension, diabetes mellitus, and dyslipidemia.
- Vitals – Stable
- Ocular examination :



HISTORY

1. Monocular vs binocular?
2. Intermittent/continuous?
3. Horizontal/vertical/torsional?
4. Increase in any direction of gaze/head tilts?
5. Greater for distance or near?


A
Left
Right


B
Left
Right

DIPLOPIA

Monocular

- photoreceptors in diff area of retina stimulated

eg: (1) corneal irregularities (astigmatism, keratoconus, dystrophies)

(2) lens irregularities

Resolves on viewing a diplopic target thru pinhole

↓

eliminates retinal perception of abN diffracted light

Binocular

↓

disconjugate eye movements as eyes are not aligned

Approach -

(1) pattern of eye movt abN

(2) structure involved

(3) etiology

Anatomic framework

(1) supranuclear

(2) nuclear

(3) fascicular

(4) subarach space

(5) orbital apex

(6) orbit

(7) NMJ

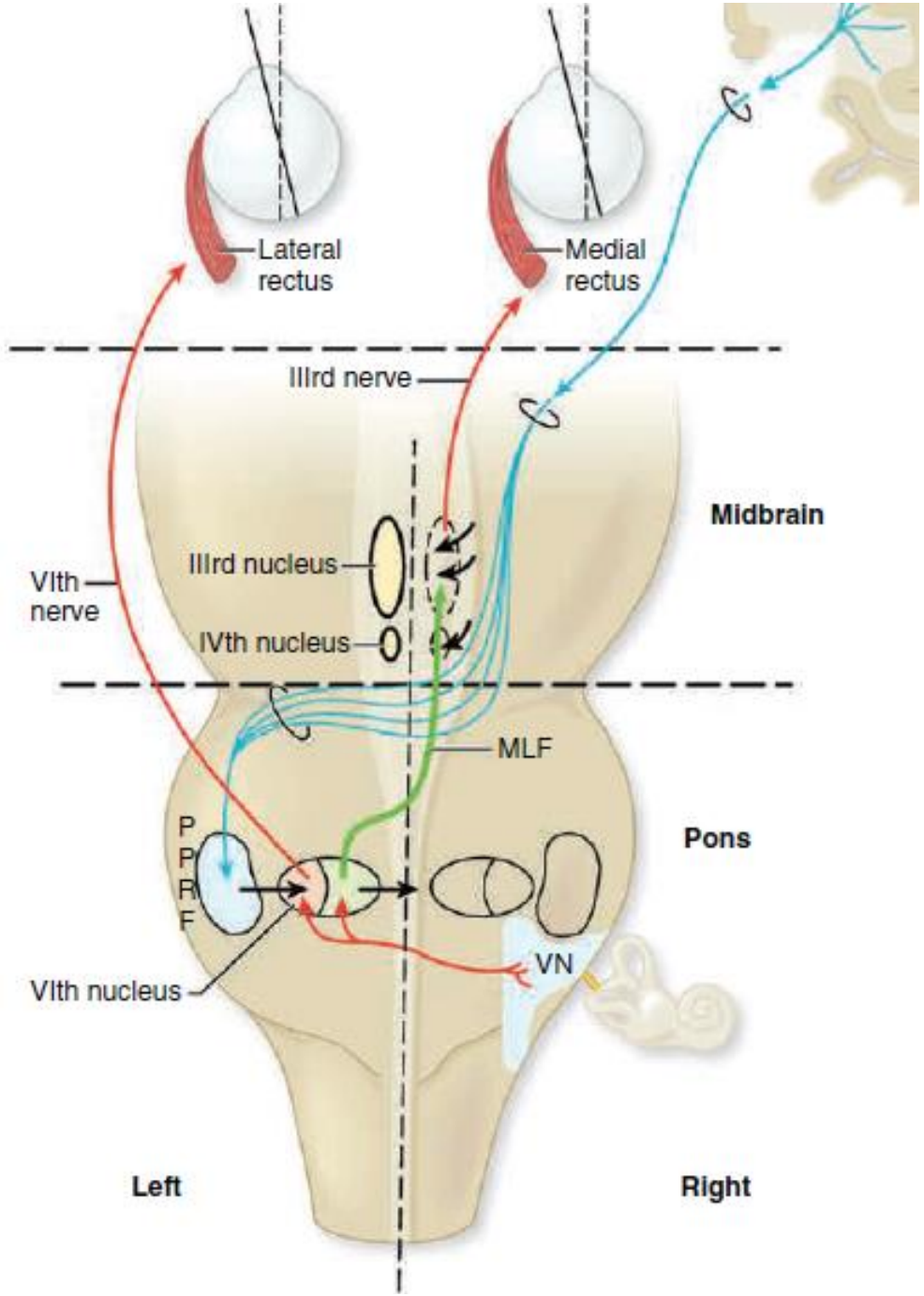
(8) EOM

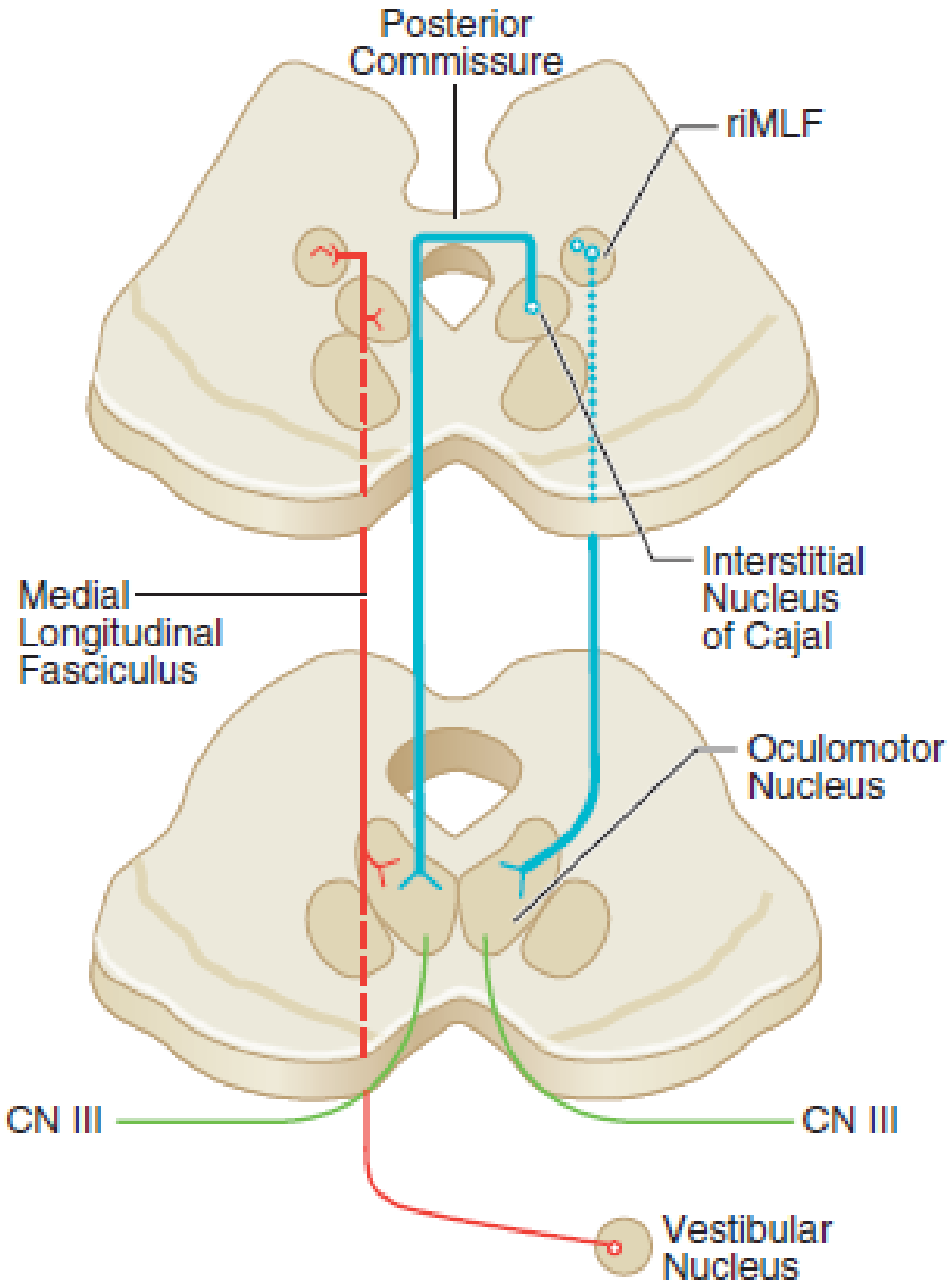
(9) ext abN of orbit



CONTROL OF GAZE

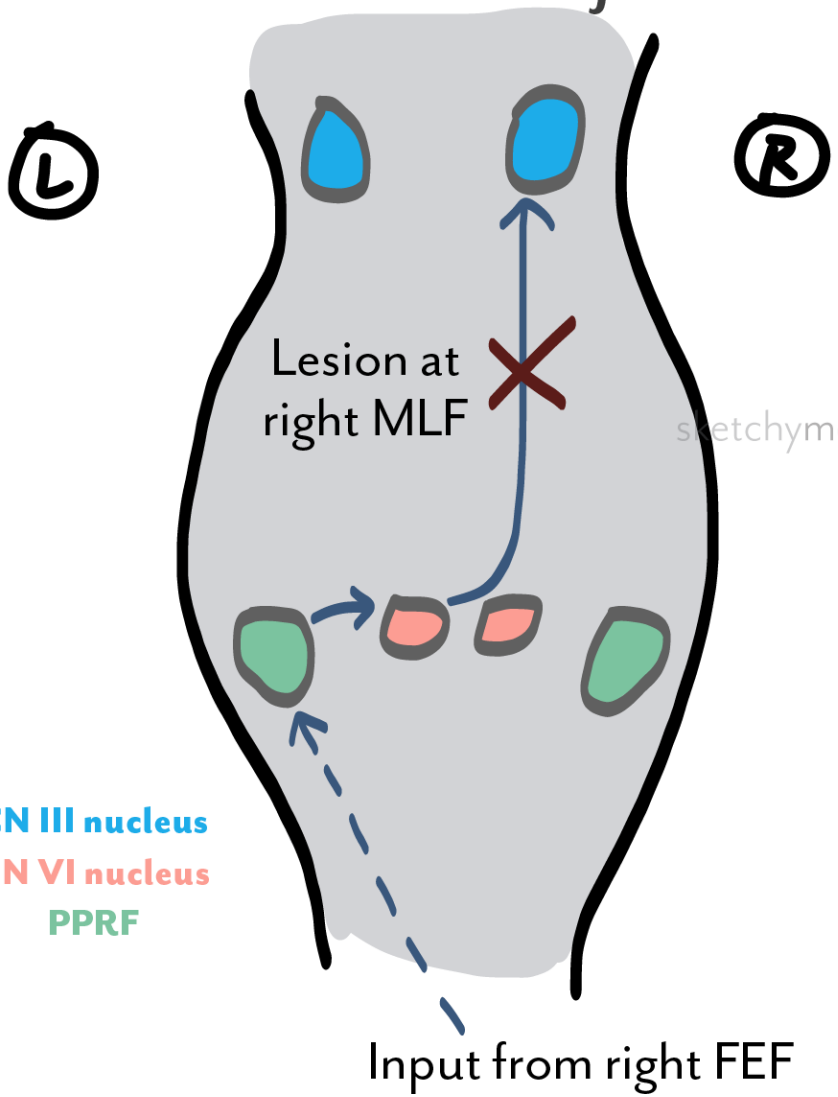
- UMN : Supranuclear , Internuclear
- LMN :
  Nuclear
  3,4,6 cranial nerve
  Neuromuscular junction
  Muscle


Lateral rectus
Medial rectus
IIIrd nerve
Midbrain
IIIrd nucleus
VIth nerve
IVth nucleus
MLF
P
P
R
F
Pons
VN
VIth nucleus
Left
Right


Posterior
Commissure
riMLF
Interstitial
Nucleus
of Cajal
Medial
Longitudinal
Fasciculus
Oculomotor
Nucleus
CN III
CN III
Vestibular
Nucleus

Internuclear Opthalmoplegia

"Look to the left"

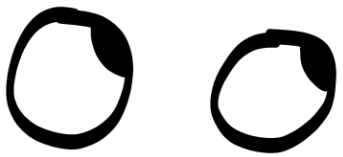


sketchymedicine.com

"Look straight ahead"



"Look to the right"



Right eye able to abduct

"Look to the left"



Right eye *unable* to adduct

"Cross your eyes"



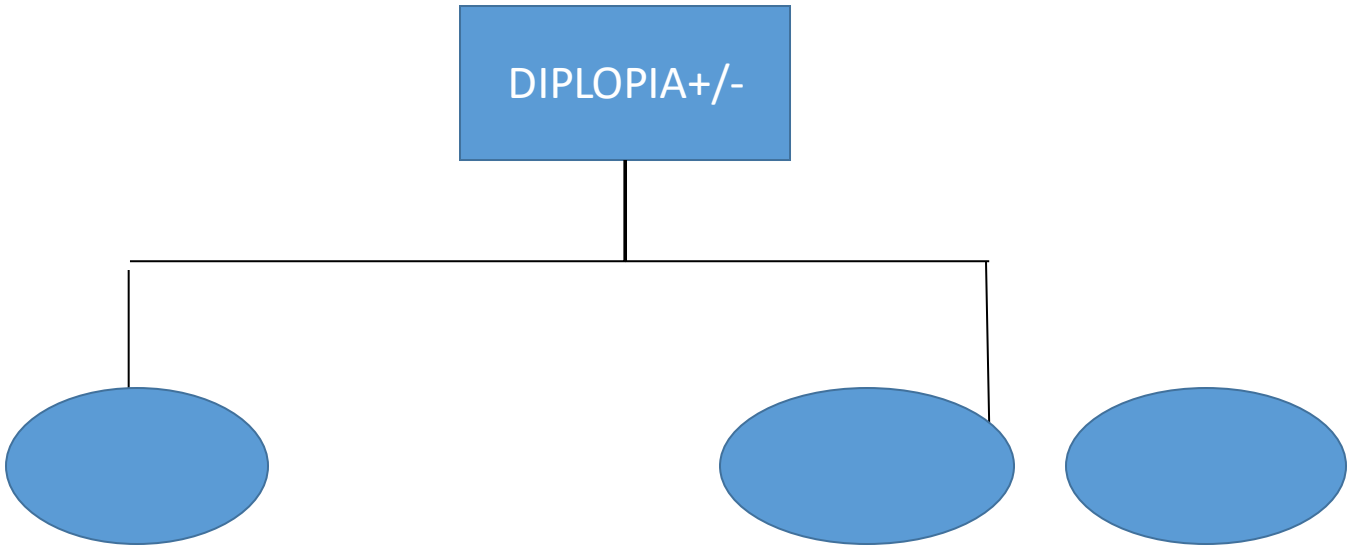
Right eye able to adduct

ASK 4 QUESTIONS

1. Monocular vs Binocular
2. Associated Neurologic symptoms or signs
3. Isolated cranial nerve palsy vs complex motility disorder
4. Screen for Giant cell arteritis in patients above 55

BINOCULAR DIPLOPIA

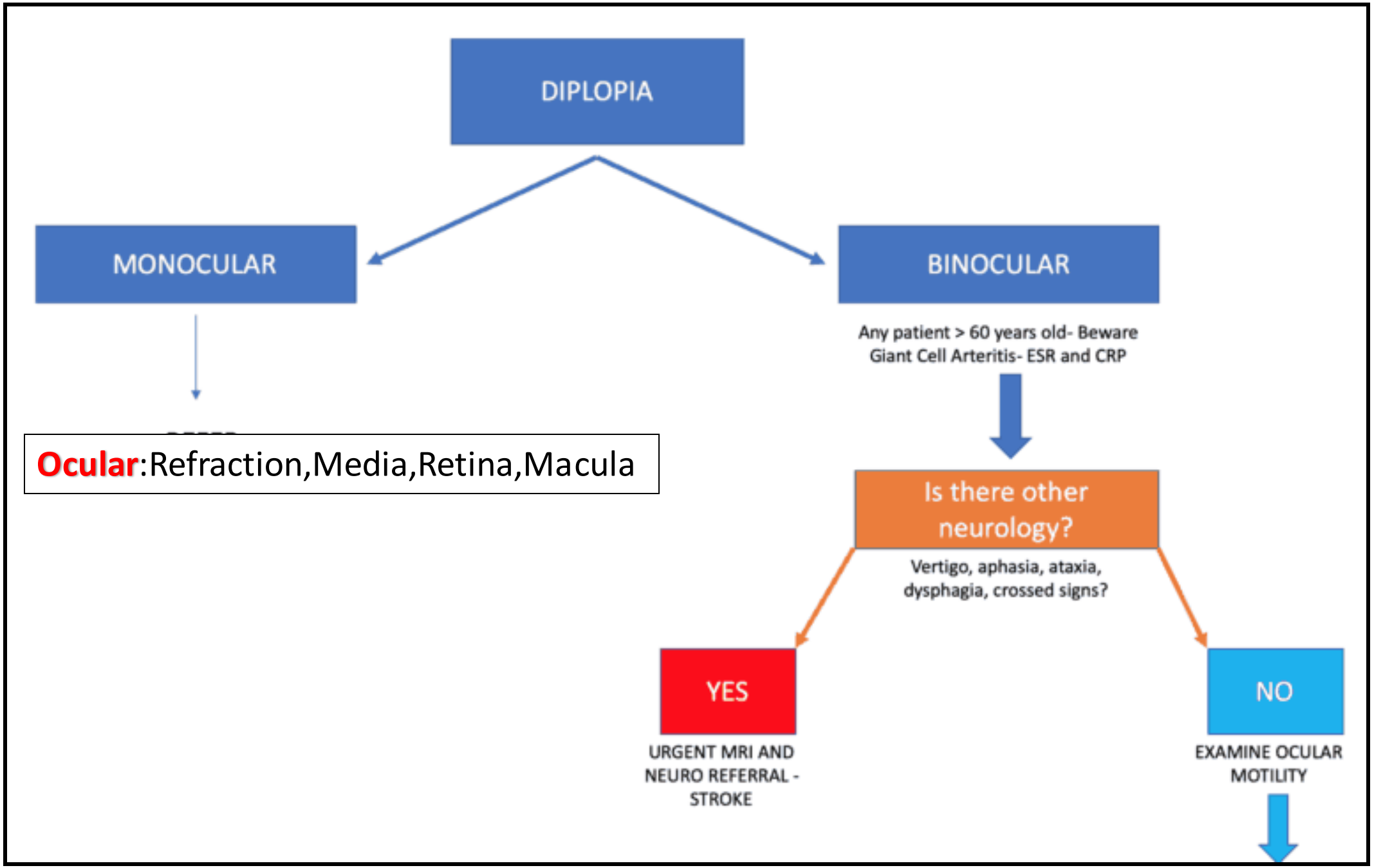
- Resolves with one eye closed
- Suggests Ocular misalignment due to issue with CNS , NMJ, Muscle
- Neurologic signs :Consider stroke,Pituitary apoplexy,

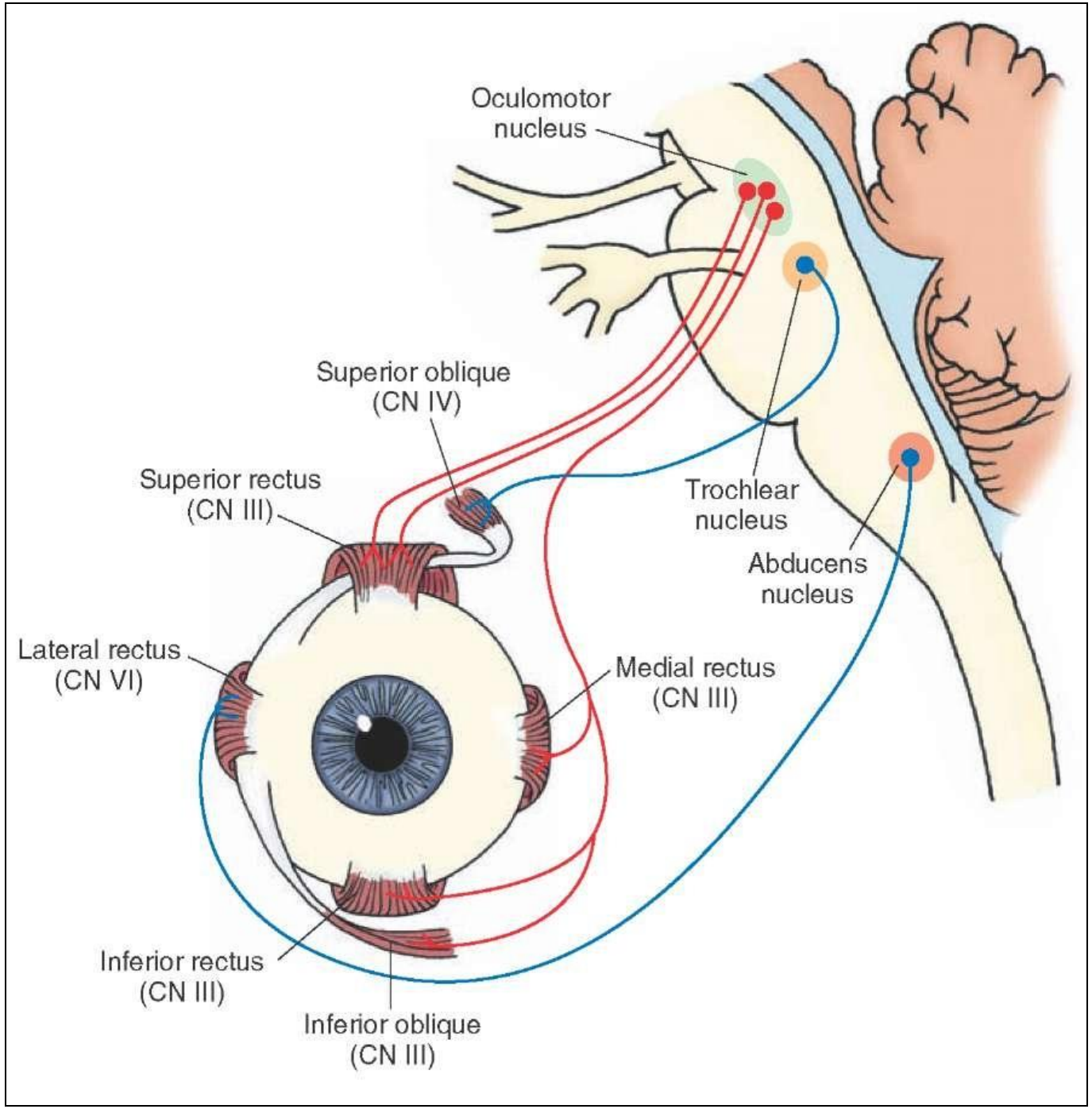


MONOCULAR
(When viewed with one eye)

BINOCULAR
(When viewed with both eye)

Ocular :Refraction,Media,Retina,Macula


DIPLOPIA
MONOCULAR
BINOCULAR
Ocular:Refraction,Media,Retina,Macula
Any patient > 60 years old- Beware Giant Cell Arteritis- ESR and CRP
Is there other neurology?
Vertigo, aphasia, ataxia, dysphagia, crossed signs?
YES
URGENT MRI AND NEURO REFERRAL - STROKE
NO
EXAMINE OCULAR MOTILITY


Oculomotor nucleus
Superior oblique (CN IV)
Superior rectus (CN III)
Trochlear nucleus
Abducens nucleus
Lateral rectus (CN VI)
Medial rectus (CN III)
Inferior rectus (CN III)
Inferior oblique (CN III)

PARK 3 STEP TEST

EXAMINE OCULAR MOTILITY

CN III PALSY

URGENT CT/CTA

Concern for compressive lesion such as aneurysm

Beware
Dilated pupil alone is NOT due to CNIII lesion
Pain does NOT assist in the diagnosis

CN VI PALSY

BILATERAL

URGENT CT

CONCERN FOR BLEED/TUMOUR
Will usually have other neurology

UNILATERAL

IS THERE TRAUMA?

YES
URGENT CT
Petrous apex fracture?

NO

IS THERE PAPILLOEDEMA?

YES
URGENT CT/CTV
CONCERN FOR RAISED ICP SECONDARY TO TUMOUR OR DURAL SINUS THROMBOSIS

NO
Causes include microangiopathy, demyelination and rarely tumour
Need MRI but not urgently

CN IV PALSY

IS THERE TRAUMA?

YES
URGENT CT

NO
Most common cause is microangiopathy. Refer Opthalmology

COMPLEX MULTIPLE CN PALSIES

CONSIDER LESIONS AT:
- Cavernous sinus
- Orbital Apex

IS THERE A HEADACHE?

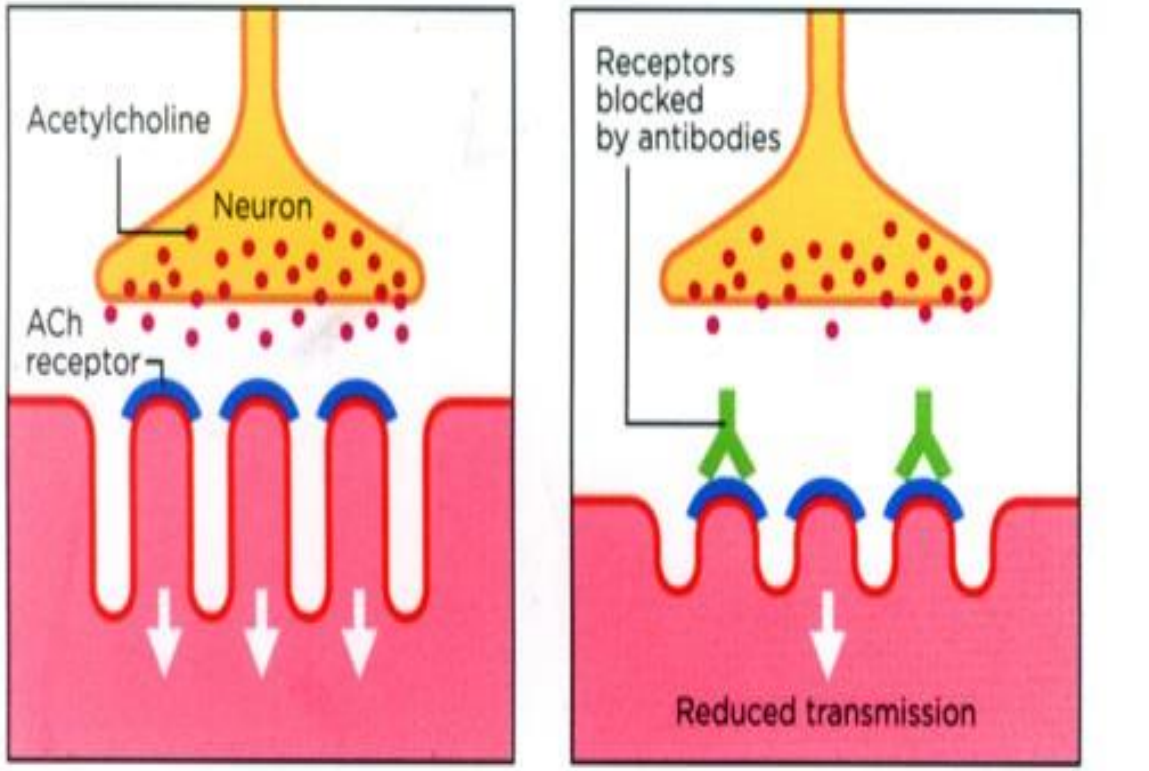
YES
URGENT CT

NO
Refer Opthalmology

MILLER FISCHER:Areflexia,Diplopia

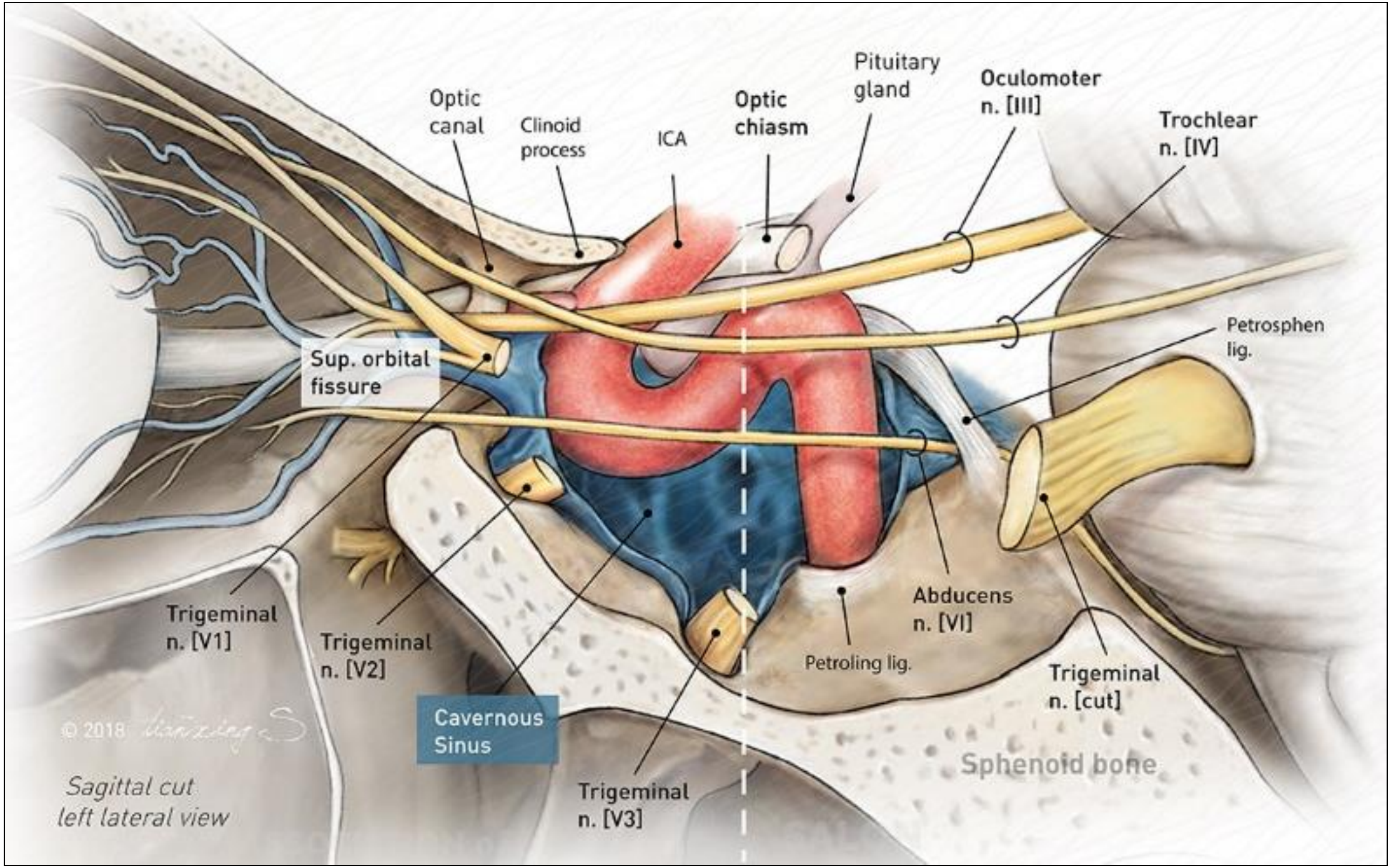
NMJ

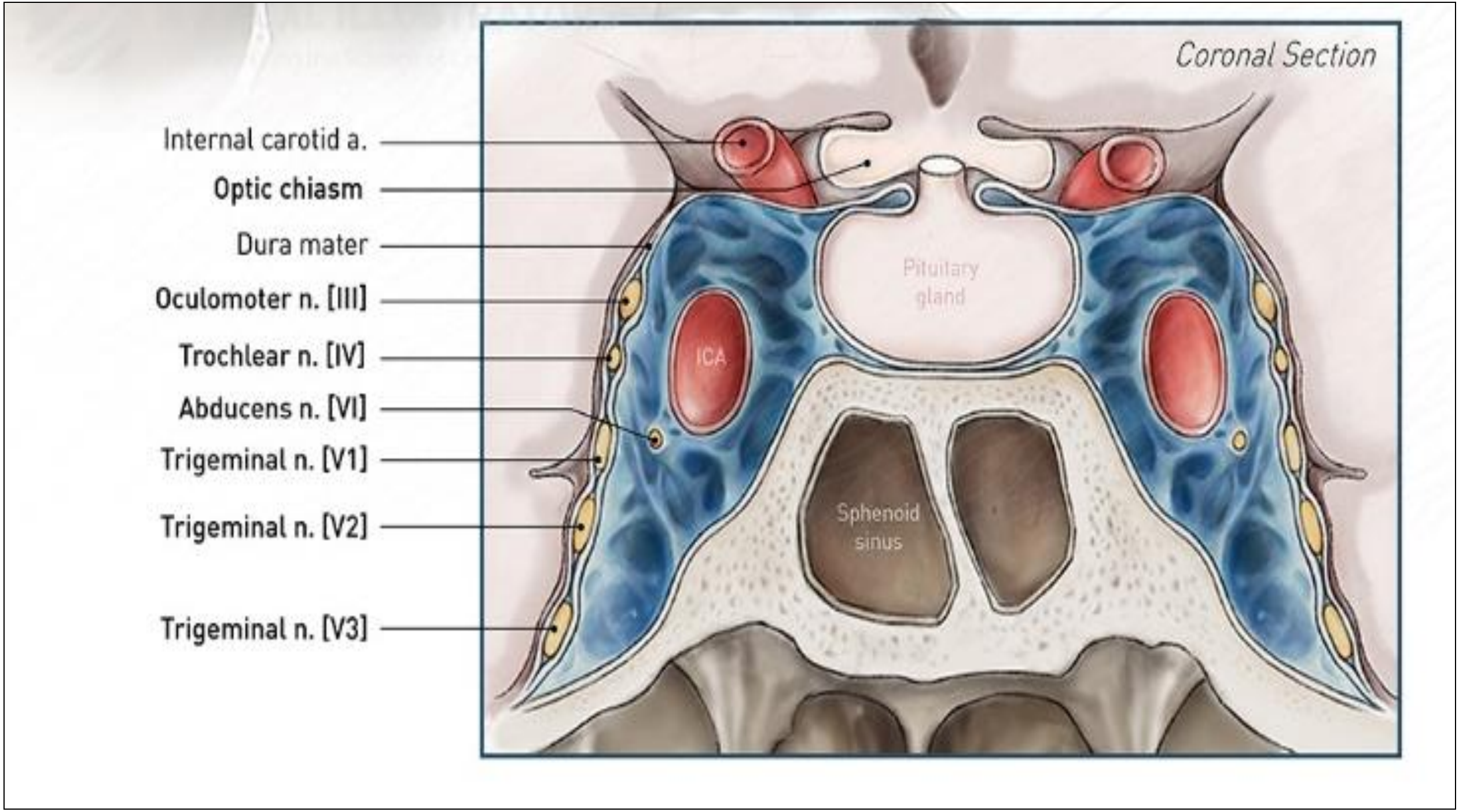
- Myasthenia Gravis
- Botulism
- Envenomation



Normal neuromuscular junction

Neuromuscular junction in myasthenia gravis


Optic canal
Clinoid process
ICA
Optic chiasm
Pituitary gland
Oculomoter n. [III]
Trochlear n. [IV]
Petrosphen lig.
Sup. orbital fissure
Trigeminal n. [V1]
Trigeminal n. [V2]
Trigeminal n. [V3]
Cavernous Sinus
Abducens n. [VI]
Petroling lig.
Trigeminal n. [cut]
Sphenoid bone
© 2018
Sagittal cut
left lateral view


Coronal Section
Internal carotid a.
Optic chiasm
Dura mater
Oculomoter n. [III]
Trochlear n. [IV]
Abducens n. [VI]
Trigeminal n. [V1]
Trigeminal n. [V2]
Trigeminal n. [V3]
Pituitary gland
ICA
Sphenoid sinus

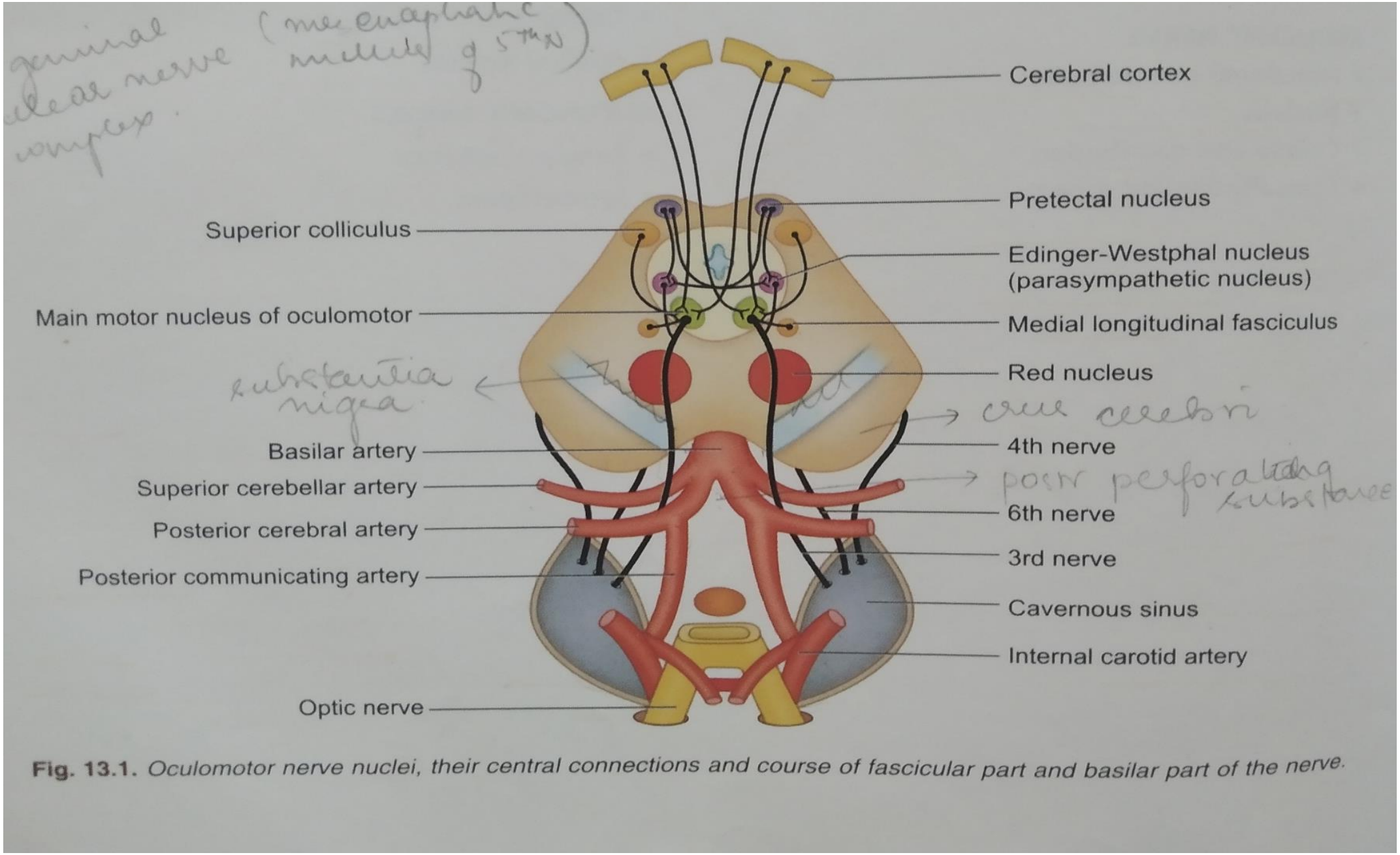


Fig. 13.1. *Oculomotor nerve nuclei, their central connections and course of fascicular part and basilar part of the nerve.*

PUPIL

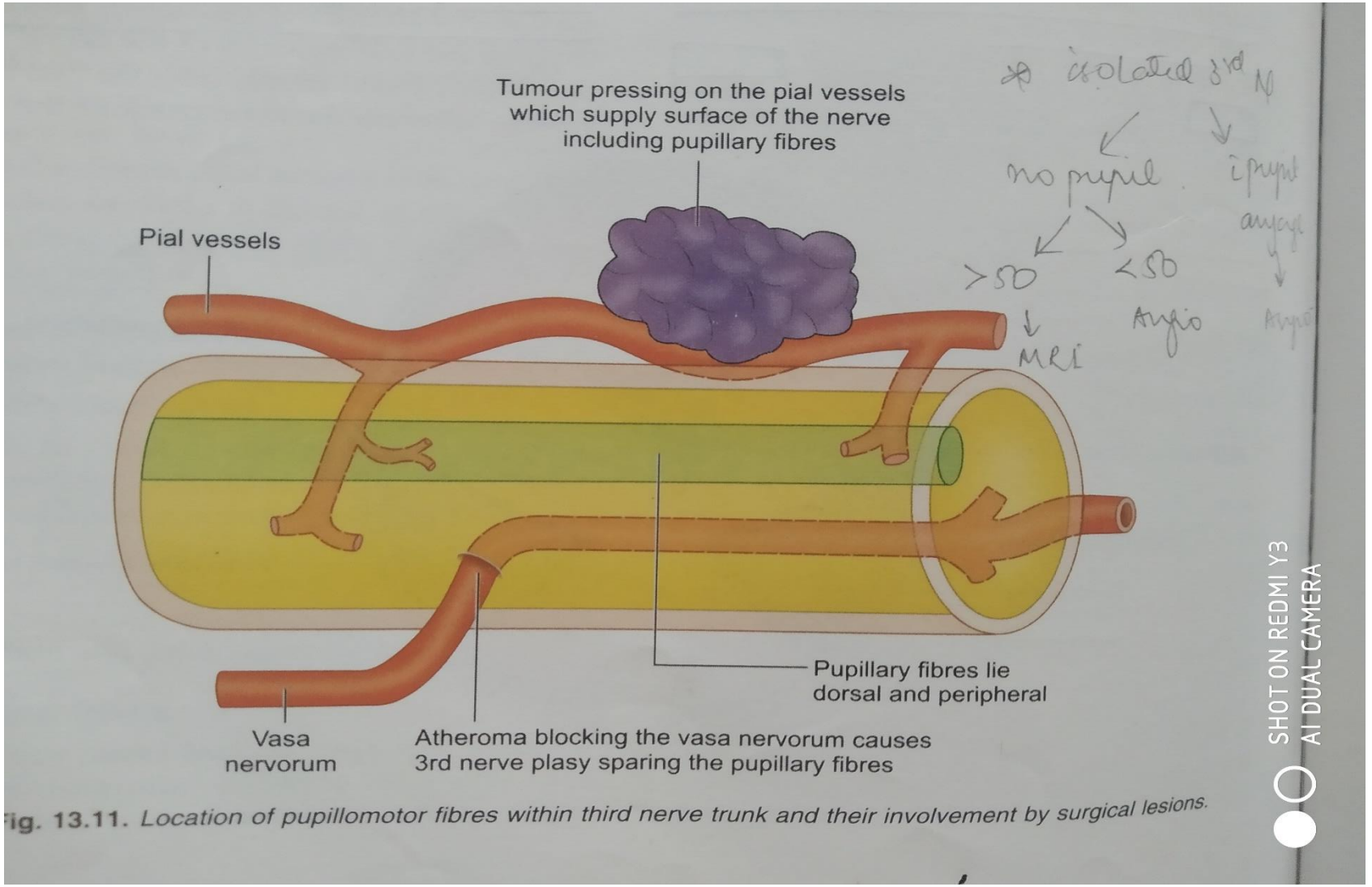


Fig. 13.11. *Location of pupillomotor fibres within third nerve trunk and their involvement by surgical lesions.*

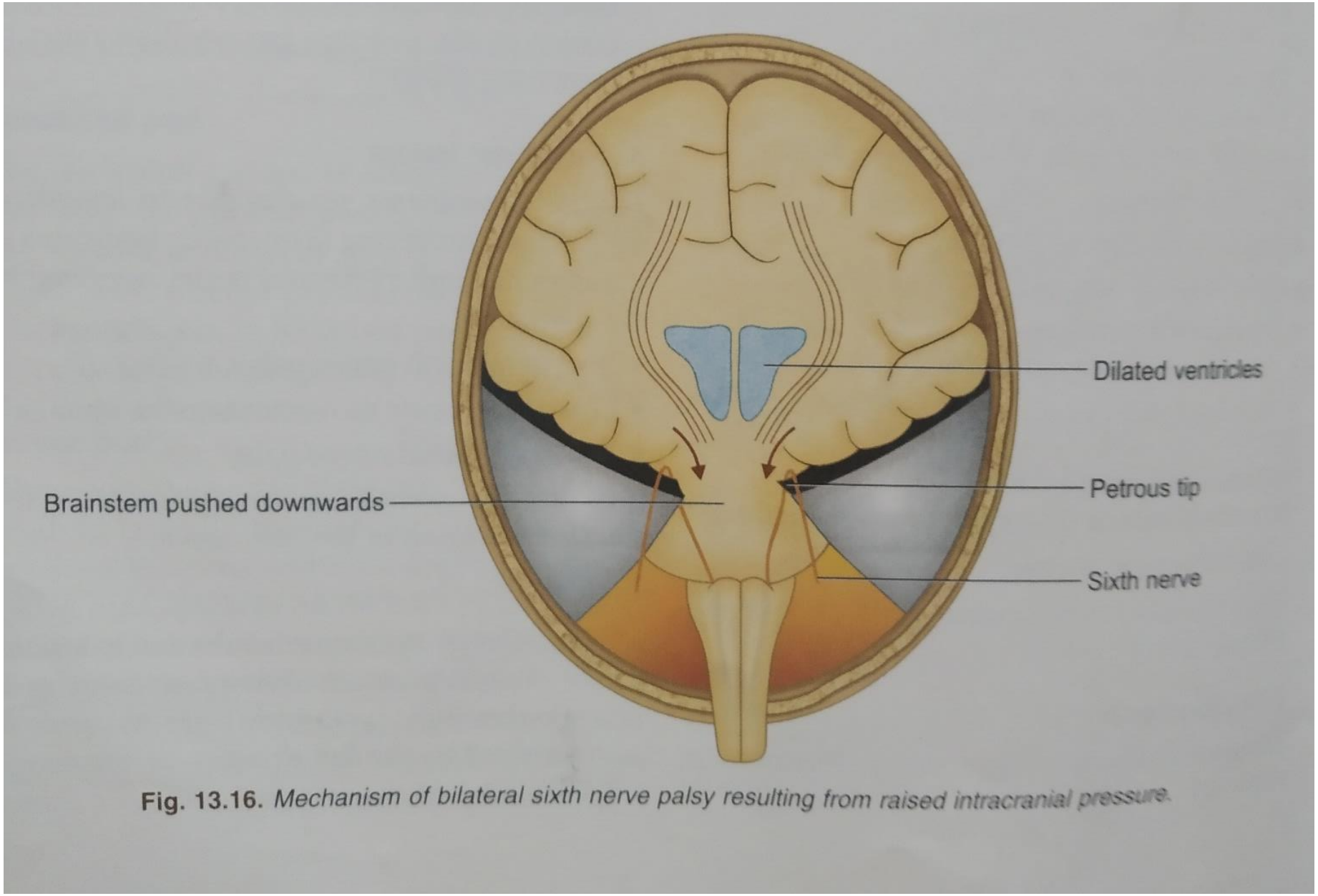


Fig. 13.16. *Mechanism of bilateral sixth nerve palsy resulting from raised intracranial pressure.*

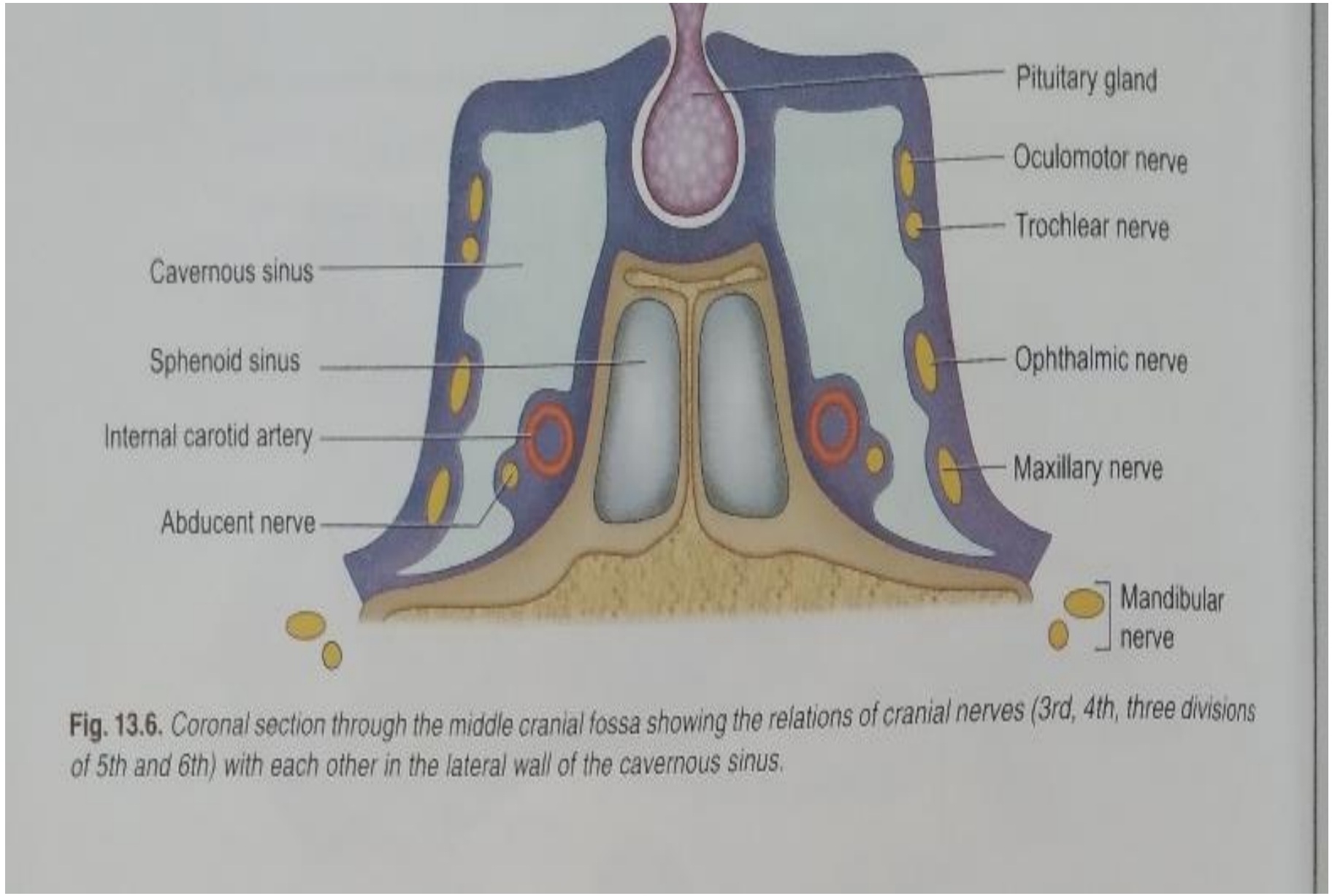


Fig. 13.6. *Coronal section through the middle cranial fossa showing the relations of cranial nerves (3rd, 4th, three divisions of 5th and 6th) with each other in the lateral wall of the cavernous sinus.*

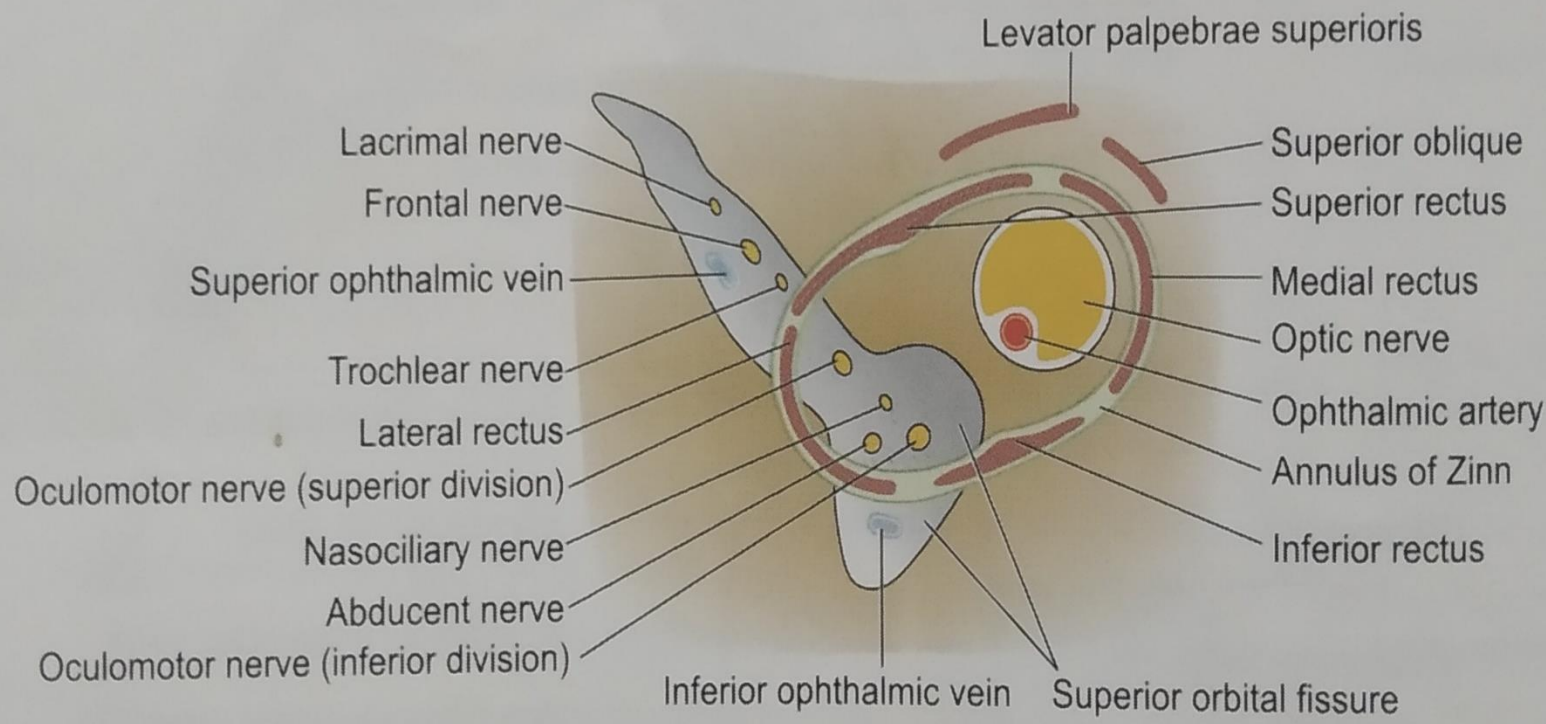
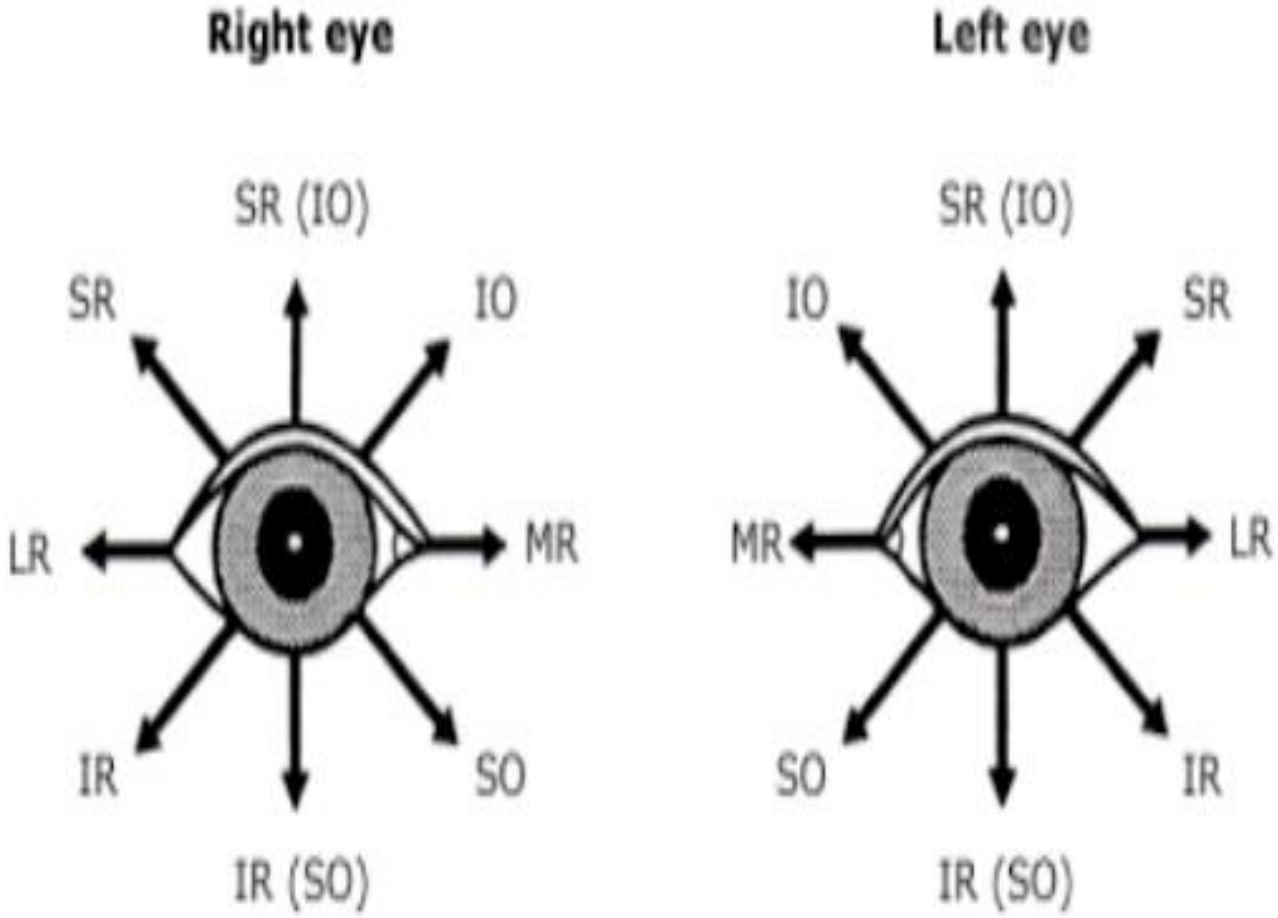
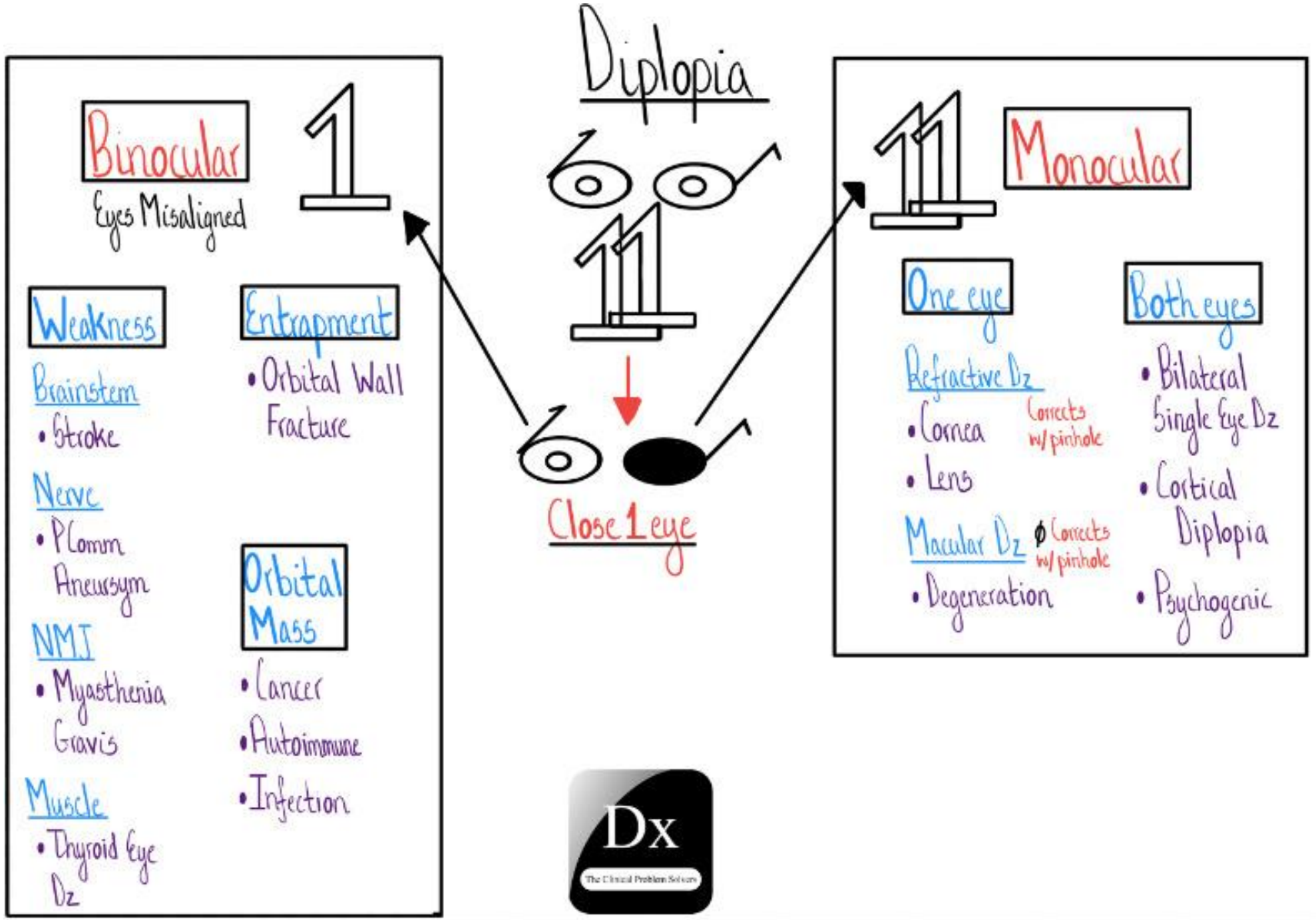


Fig. 13.7. *Apical part of the orbit showing the origin of extraocular muscles, the common tendinous ring and the structures passing through the superior orbital fissure.*

MUSCLE/ MECHANICAL



- Thyroid
- Trauma
- Tumor


Diplopia
Binocular
Eyes Misaligned
Weakness
Brainstem
• Stroke
Nerve
• PComm Aneurysm
NMJ
• Myasthenia Gravis
Muscle
• Thyroid Eye Dz
Entrapment
• Orbital Wall Fracture
Orbital Mass
• Cancer
• Autoimmune
• Infection
Close 1 eye
Monocular
One eye
Refractive Dz
• Cornea
• Lens
Corrects w/ pinhole
Macular Dz
• Degeneration
ϕ Corrects w/ pinhole
Both eyes
• Bilateral Single Eye Dz
• Cortical Diplopia
• Psychogenic
Dx
The Clinical Problem Solvers

Summarise!!


thank you
thank you
thank you
thank you
thank you
thank you
thank you
thank you